

Photosynthesis in Plants

Overview

Plants capture light energy and store it as chemical energy.

Photosynthesis powers nearly all life on Earth.

It releases oxygen into the atmosphere.

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Light Absorption

Chlorophyll absorbs mostly red and blue light.

Accessory pigments broaden the range of usable wavelengths.

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Light Reactions

Light energy drives the transfer of electrons along a chain.
The energy is stored as ATP and NADPH for later use.

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Calvin Cycle

The Calvin cycle fixes carbon dioxide into sugar.
It runs in the stroma of the chloroplast.

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Photolysis

Water is split during photolysis releasing oxygen.

The electrons from water replace those lost from the reaction centre.

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Limiting Factors

Light intensity, temperature and carbon dioxide all limit the rate.
A farmer can raise yield by raising the most limiting factor.

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Net photosynthesis stops when the lights are turned off.
At night respiration uses the stored sugars for energy.

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Net productivity depends on light intensity and temperature.

Above a certain temperature enzymes denature and output drops.